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Uniform Inference in Predictive Regression Models

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The restricted likelihood has been found to provide a well-behaved likelihood ratio test in the predictive regression model even when the regressor variable exhibits almost unit root behavior. Using the weighted least squares approximation to the restricted likelihood obtained in Chen and Deo, we provide a quasi restricted likelihood ratio test (QRLRT), obtain its asymptotic distribution as the nuisance persistence parameter varies, and show that this distribution varies very slightly. Consequently, the resulting sup bound QRLRT is shown to maintain size uniformly over the parameter space without sacrificing power. In simulations, the QRLRT is found to deliver uniformly higher power than competing procedures with power gains that are substantial.

KEY WORDS: Efron curvature; REML; Sup bound.

1. INTRODUCTION

The basic predictive regression model is given by

$$\begin{aligned} Y_t &= \eta + \beta X_{t-1} + u_t \\ X_t &= \mu + \alpha X_{t-1} + v_t \end{aligned} \quad \begin{pmatrix} u_t \\ v_t \end{pmatrix} \sim \text{iid } N(0, \Sigma), \quad (1.1)$$

where $X_0 = 0$ and $\alpha \in (0, 1]$. In this model, it is well known that carrying out inference on β using standard procedures, such as t -statistics or the usual likelihood ratio test, is problematic, particularly when the nuisance autoregressive parameter α in the regressor X_t is close to unity, as is often the case in empirical applications. Chen and Deo (2009a) showed that the reason why the standard inference procedures fail is primarily due to the presence of the nuisance intercept parameter μ and not as much due to the autoregressive parameter α , as was generally thought. Hence, they proposed using the restricted likelihood (RL), which is the exact likelihood of any linear transformation of the data (Y_t, X_t) that eliminate the intercepts η and μ (the linear transformations are not unique, though the RL is unique up to a multiple). Chen and Deo (2009a) showed that the RL possesses low Efron curvature, due to which it yields a well behaved restricted likelihood ratio test (RLRT). When the autoregressive parameter α is in the stationary region, they show that the error in the χ^2 distribution approximation to that of the RLRT is very small and, more importantly, does not depend on α , that is, the distribution of the RLRT is second-order pivot with respect to α . As a consequence, the deviation of the RLRT distribution from χ^2 would not be expected to be substantial as α approached unity, an expectation that was supported in their simulations. This finding suggests that a sup-bound approach to inference for β based on the distribution of the RLRT across different configurations of α will result in a testing procedure that, while maintaining correct size, will not result in a substan-

tial power loss. However, Chen and Deo (2009a) did not obtain the limiting distribution of the RLRT when α is close to the unit root. As a consequence, the current theory does not allow for uniform inference over the entire parameter space of α .

In this article, we consider the weighted least squares approximate restricted likelihood (WLSRL) of vector autoregressive (VAR) processes (Chen and Deo 2010) which approximates the exact RL and has the virtue of yielding an explicit weighted least squares estimator of the AR coefficient matrix. Chen and Deo (2010) showed that this WLSRL estimator is asymptotically equivalent to the exact RL maximum likelihood estimator and shares its superior bias properties. We obtain the limiting distribution of the WLSRL estimators of the predictive regression model (1.1) under three scenarios for the AR coefficient α , viz. stationarity, moderate deviations from unity and local to unity. We then obtain the asymptotic distribution of the resulting quasi-RLRT (QRLRT), based on the WLSRL, under these scenarios. Since the WLSRL approximates the RL, the discussion above suggests that this limiting distribution also should not vary too much as α varies. Indeed, we find in simulations that the resulting sup-bound test based on the QRLRT maintains size over the entire parameter space without significant power loss and has uniformly higher power than the Jansson Moreira (2006) test, with power gains that can be substantial. The asymptotic distribution of the QRLRT is also obtained under local alternatives allowing for varying degrees of persistence in the autoregressive parameter.

In the next section, we introduce the WLSRL for the predictive regression model and obtain the limiting distributions of

the resulting WLSRL estimators, comparing them to the limiting distributions of the usual ordinary least squares (OLS) estimators. In Section 3, we obtain the limiting distribution of the QRLRT under the three scenarios mentioned above for α and define the sup-bound test procedure for inference on β , which yields uniform inference over the entire nuisance parameter space. We also obtain the limiting distribution of the QRLRT under various specifications of local alternatives for β . Finally, in Section 4 we evaluate finite sample performance of the sup-bound QRLRT test through a simulation study and compare its performance to the procedure of Jansson and Moreira (2006). All proofs are relegated to the appendix at the end.

2. WEIGHTED LEAST SQUARES APPROXIMATED LIKELIHOOD

Though the RL of AR processes is known to have good properties (see Chen and Deo 2009a, 2009b, 2012), it is difficult to use it in vector AR series, since optimizing the RL is not easy due to its nonlinear nature. Hence, Chen and Deo (2010) obtained a weighted least squares approximation to the RL of vector AR processes, which yields weighted least squares estimators that are easy to compute and which share the good properties of the exact RL estimators. Letting $Z_t = (Y_t, X_t)'$, the predictive regression model (1.1) implies that Z_t is a bivariate AR process, allowing us to use Chen and Deo (2010) to obtain the WLSRL of Z_t , as follows.

We start by defining some notation. For $t = 2, \dots, n$ and any time series ξ_t , we will denote the corresponding sample mean corrected series and its lag series as,

$$\xi_{t,\hat{\mu}} = \xi_t - \frac{1}{n-1} \sum_{u=2}^n \xi_u, \quad \xi_{t-1,\hat{\mu}} = \xi_{t-1} - \frac{1}{n-1} \sum_{u=2}^n \xi_{u-1}.$$

Furthermore, denote the series with the first observation subtracted as

$$\xi_{s,d} = \xi_s - \xi_1, \quad s = 1, \dots, n.$$

For $t = 2, \dots, n$, let

$$\epsilon_{t,\hat{\mu}}(\beta, \alpha) = \begin{pmatrix} u_{t,\hat{\mu}} \\ v_{t,\hat{\mu}} \end{pmatrix} = \begin{pmatrix} Y_{t,\hat{\mu}} - \beta X_{t-1,\hat{\mu}} \\ X_{t,\hat{\mu}} - \alpha X_{t-1,\hat{\mu}} \end{pmatrix} \quad \text{and} \\ r_t(\beta, \alpha) = \begin{bmatrix} Y_{t,d} - \beta X_{t-1,d} \\ X_{t,d} - \alpha X_{t-1,d} \end{bmatrix}.$$

Let $(\hat{\Sigma}, \hat{\beta}, \hat{\alpha})$ be any consistent estimators of (Σ, β, α) . For example, such consistent estimators can be obtained from OLS estimation of model (1.1). From Lemma 1 of Chen and Deo (2010), the WLSRL of model (1.1) based on the observations $(Y_t, X_t)'$, $t = 1, \dots, n$ is given by

$$Q_n(\beta, \alpha) = Q_{\text{OLS}}(\beta, \alpha) + Q_W(\beta, \alpha), \quad (2.1)$$

where

$$Q_{\text{OLS}}(\beta, \alpha) = \sum_{t=2}^n \epsilon'_{t,\hat{\mu}}(\beta, \alpha) \hat{\Sigma}^{-1} \epsilon_{t,\hat{\mu}}(\beta, \alpha), \\ Q_W(\beta, \alpha) = \frac{1}{n-1} \left\{ \sum_{s=2}^n r_s(\beta, \alpha) \right\}' \hat{W}_n \left\{ \sum_{t=2}^n r_t(\beta, \alpha) \right\},$$

and $\hat{W}_n = W_n(\hat{\Sigma}, \hat{\beta}, \hat{\alpha})$ with

$$W_n(\Sigma, \beta, \alpha) = \{\Sigma + (n-1)(I - H)\Sigma(I - H')\}^{-1}, \\ H = \begin{pmatrix} 0 & \beta \\ 0 & \alpha \end{pmatrix}. \quad (2.2)$$

The objective function Q_{OLS} is the usual least squares function which uses the sample mean corrected data. The second term, Q_W , in the WLSRL function $Q_n(\alpha, \beta)$ serves as a correction, with the magnitude of correction depending on its weight function W_n , which in turn depends on α . Lemma 2 in the Appendix shows how this weight function W_n changes as $\alpha = 1 - c/k_n$ varies. When $k_n \ll \sqrt{n}$, the sample mean $\bar{X}$ is a consistent estimator of μ (Giraitis and Phillips 2012) and so estimators based on sample mean corrected data will behave well. In this situation, as Lemma 2 shows, the weight function W_n is asymptotically negligible and the WLSRL behaves approximately like the OLS function. However, when $k_n \gg \sqrt{n}$, the sample mean is not a consistent estimator of μ (Giraitis and Phillips 2012) but the weight function W_n is nonnegligible. As a matter of fact, it results in a correction term Q_W which is such that the WLSRL estimator of α is asymptotically the same as the OLS estimator computed with $X_{t,d} = X_t - X_1$, and so achieves location invariance without depending on the sample mean.

Letting $Z_t = (Y_t, X_t)'$, the WLSRL estimator of $(\beta, \alpha)'$ is the minimizer of $Q_n(\beta, \alpha)$ and is given by

$$(\hat{\beta}_{\text{WLS}}, \hat{\alpha}_{\text{WLS}})' \\ = \left[\sum_{t=2}^n X_{t-1,\hat{\mu}}^2 \hat{\Sigma}^{-1} + \frac{1}{n-1} \left(\sum_{t=2}^n X_{t-1,d} \right)^2 \hat{W}_n \right]^{-1} \\ \times \left[\hat{\Sigma}^{-1} \sum_{t=2}^n X_{t-1,\hat{\mu}} Z_{t,\hat{\mu}} + \frac{1}{n-1} \hat{W}_n \sum_{s=2}^n X_{s-1,d} \sum_{t=2}^n Z_{t,d} \right]. \quad (2.3)$$

The next theorem gives the limiting distribution of the WLSRL estimator. The parameter $\rho = \text{Corr}(u_t, v_t)$ will play an important role in all of our theoretical results.

Theorem 1. Under the model in (1.1), the WLSRL estimator given in (2.3) has the following limiting distributions:

(i) If $\alpha \in (-1, 1)$ and α is fixed,

$$\sqrt{n} \begin{pmatrix} \hat{\beta}_{\text{WLS}} - \beta \\ \hat{\alpha}_{\text{WLS}} - \alpha \end{pmatrix} \xrightarrow{D} \mathcal{N}(0, (1 - \alpha^2)\Sigma).$$

(ii) If $\alpha = 1 - c/k_n$, where $k_n = n^\lambda$, $\lambda \in (0, 1)$, then

$$\sqrt{nk_n} \begin{pmatrix} \hat{\beta}_{\text{WLS}} - \beta \\ \hat{\alpha}_{\text{WLS}} - \alpha \end{pmatrix} \xrightarrow{D} \mathcal{N}(0, 2c\Sigma).$$

(iii) If $\alpha = 1 - c/n$,

$$n \begin{pmatrix} \hat{\beta}_{\text{WLS}} - \beta \\ \hat{\alpha}_{\text{WLS}} - \alpha \end{pmatrix} \xrightarrow{D} \begin{pmatrix} \sqrt{1 - \rho^2} & \rho \\ 0 & 1 \end{pmatrix} \\ \times \begin{pmatrix} \int J_{c,\mu}^2(r) dr \\ \int J_c^2(r) dr \end{pmatrix}^{-1/2} \mathcal{Z},$$

where

$$J_c(r) = \int_0^1 e^{-(r-u)c} dW(u), \quad J_{c,\mu}(r) = J_c(r) - \int_0^r J_c(s) ds$$

and $W(r)$ is Brownian Motion that is independent from $\mathcal{Z} \sim N(0, 1)$.

The result above shows that regardless of the value of α , the limiting distribution of the WLSRL estimator of α is identical to that of the OLS estimator of α in a model without intercept. It is well known that when α is close to unity, this distribution is markedly different from that of the OLS estimator in the model with an intercept, with significantly smaller bias and variance. In the next section, we study the QRLRT for testing the hypothesis that $\beta = 0$.

3. THE QUASI-RESTRICTED LIKELIHOOD RATIO TEST

As pointed out in the Introduction, the WLSRL approximates the exact RL of a vector AR process, and hence can be used to construct a quasi-RL ratio test to test any hypothesis on the parameters in the coefficient matrix. In the predictive regression context, one is interested in testing the null hypothesis of no predictability, that is, in testing $H_0 : \beta = 0$. To compute the corresponding QRLRT for this hypothesis, one needs to first obtain the constrained estimate of the nuisance parameter α under the hypothesis $\beta = 0$. This constrained estimator is the minimizer of $Q_n(0, \alpha)$ and is given by

$$\begin{aligned} \hat{\alpha}_{\text{WLS}}^0 &= \left\{ \hat{\sigma}^{22} \sum_{t=2}^n X_{t-1,\hat{\mu}}^2 + \frac{\hat{w}_{22}}{n-1} \left(\sum_{t=2}^n X_{t-1,d} \right)^2 \right\}^{-1} \\ &\times \left\{ \sum_{t=2}^n X_{t-1,\hat{\mu}} (\hat{\sigma}^{21} Y_{t,\hat{\mu}} + \hat{\sigma}^{22} X_{t,\hat{\mu}}) + \frac{1}{n-1} \right. \\ &\times \left. \left(\sum_{s=2}^n X_{s-1,d} \right) \left(\sum_{t=2}^n \hat{w}_{21} Y_{t,d} + \hat{w}_{22} X_{t,d} \right) \right\}, \quad (3.1) \end{aligned}$$

where $\hat{\sigma}^{ij}$ and $\hat{w}_{ij}$ are the (i, j) th entries of $\hat{\Sigma}^{-1}$ and $\hat{W}_n$, respectively. The following lemma provides the asymptotic distribution of this constrained estimator when α lies in different parts of the parameter space.

Lemma 1. When $\beta = 0$, we get

- (i) If $\alpha \in (-1, 1)$ and α is fixed,
- (ii) If $\alpha = 1 - c/k_n$, where $k_n = n^\lambda$, $\lambda \in (0, 1)$,
- (iii) If $\alpha = 1 - c/n$,

$$\begin{aligned} n(\hat{\alpha}_{\text{WLS}}^0 - \alpha) &\xrightarrow{D} \frac{(1 - \rho^2 g_{c,\rho}) \int J_c(\lambda) dW(\lambda)}{\int J_c^2(\lambda) d\lambda} \\ &+ \frac{\rho g_{c,\rho}^{1/2} \sqrt{1 - \rho^2 g_{c,\rho}} \mathcal{Z}}{\sqrt{\int J_c^2(\lambda) d\lambda}}, \end{aligned}$$

where $J_c(r)$, $W(r)$, and $\mathcal{Z}$ are defined as in Theorem 1 and

$$g_{c,\rho} = \frac{1 - \left(\int J_c^2(\lambda) d\lambda \right)^{-1} \left(\int J_c(\lambda) d\lambda \right)^2}{1 - \rho^2 \left(\int J_c^2(\lambda) d\lambda \right)^{-1} \left(\int J_c(\lambda) d\lambda \right)^2}.$$

A QRLRT statistic for testing $H_0 : \beta = 0$ based on the WLSRL may now be defined as

$$\Lambda_n = Q_n(0, \hat{\alpha}_{\text{WLS}}^0) - Q_n(\hat{\beta}_{\text{WLS}}, \hat{\alpha}_{\text{WLS}}), \quad (3.2)$$

and the next theorem provides its asymptotic distribution.

Theorem 2. For the QRLRT defined above, we have, under $H_0 : \beta = 0$,

- (i) If $\alpha \in (-1, 1)$ and α is fixed, $\Lambda_n \xrightarrow{D} \chi_1^2$.
- (ii) If $\alpha = 1 - c/k_n$, where $k_n = n^\lambda$, $\lambda \in (0, 1)$, $\Lambda_n \xrightarrow{D} \chi_1^2$.
- (iii) If $\alpha = 1 - c/n$, $\Lambda_n \xrightarrow{D} \Lambda_{c,\rho}$, where

$$\begin{aligned} \Lambda_{c,\rho} &:= \left\{ \rho g_{c,\rho}^{1/2} \tau_c + \sqrt{1 - \rho^2 g_{c,\rho}} \mathcal{Z} \right\}^2, \\ \tau_c &= \frac{\int J_c(\lambda) dW(\lambda)}{\sqrt{\int J_c^2(\lambda) d\lambda}}, \end{aligned}$$

and $\mathcal{Z} \sim \mathcal{N}(0, 1)$ is independent from τ_c and $g_{c,\rho}$ is defined as in Lemma 1.

Remark 1. By its definition, $0 < g_{c,\rho} \leq 1$ and the equality holds only if $|\rho| = 1$.

Remark 2. The random variable τ_c is the limiting distribution of the t -statistic for testing α in a local-to-unity AR(1) process with zero mean. As $c \rightarrow \infty$, $\tau_c \xrightarrow{D} \mathcal{N}(0, 1)$ (Bobkoski 1983; Phillips 1987) and, from Phillips (1987), $g_{c,\rho} \xrightarrow{p} 1$. Thus $\Lambda_{c,\rho} \xrightarrow{D} \chi_1^2$, as $c \rightarrow \infty$.

Remark 3. Larsson (1995) showed that $|P(\tau_c^2 > x) - P(\chi_1^2 > x)| = o(1)$ as $x \rightarrow \infty$. Thus the right tail of $\Lambda_{c,\rho}$ should be very close to that of a χ_1^2 .

Remark 4. Note that the limiting distribution of QRLRT in Theorem 2 above is obtained under the assumption of the null hypothesis $H_0 : \beta = 0$, which is the commonly considered hypothesis of no predictability. If one wished to test an arbitrary value of β , say $H_0 : \beta = b$ for some $b \neq 0$, then one would simply replace the Y_t values by $Y_t - bX_{t-1}$ and then test the $H_0 : \beta = 0$. Hence, there is no loss of generality in focusing on the null value of 0 for β .

The theorem above shows that the QRLRT distribution changes depending on the value of the nuisance parameter α . Hence, to have a test that controls size over the entire nuisance parameter space, one would have to work with the sup-bound critical value, with the supremum taken over all possible values of α ; see, for example, Cavanagh, Elliot, and Stock (1995). More specifically, for a given ρ and level of significance δ , define

$$\bar{q}_{\rho,\delta} = \sup_{c \geq 0} \{q_{\rho,\delta}(c) : P(\Lambda_{c,\rho} > q_{\rho,\delta}(c)) = \delta\}.$$

Table 1. $P(\Lambda_{c,\rho} > \bar{q}_{\rho,0.05})$ in percentage $\delta = 5\%$

ρ/c	0	1	3	5	7	9	χ_1^2
0.98	5%	4.66%	4.31%	4.21%	4.16%	4.16%	4.13%
0.75	5%	4.57%	4.16%	4.01%	3.94%	3.95%	3.91%
0.50	5%	4.78%	4.55%	4.47%	4.43%	4.42%	4.40%

NOTE: Based on 1,000,000 replications simulated from the distribution of $\Lambda(c, \rho)$. Each replication of $\Lambda(c, \rho)$ is generated by approximating the relevant stochastic integrals in it using 32,768 (2^{15}) observations from a Gaussian white-noise.

Since, as pointed out in Remark 2, the local-to-unity limiting distribution of the QRLRT transitions to the limiting chi-square distribution as $c \rightarrow \infty$, the results of Theorem 2 imply that

$$\lim_{n \rightarrow \infty} \sup_{\alpha} P_{\beta=0}(\Lambda_n > \bar{q}_{\rho,\delta}) \leq \delta \quad \alpha \in (-1, 1].$$

Though ρ is not known, it is trivial to obtain a consistent estimator of it and hence of $\bar{q}_{\rho,\delta}$, which we denote by $\bar{q}_{\hat{\rho},\delta}$. We thus get the following theorem:

Theorem 3. For testing $H_0 : \beta = 0$,

$$\lim_{n \rightarrow \infty} \sup_{\alpha} P_{\beta=0}(\Lambda_n > \bar{q}_{\hat{\rho},\delta}) \leq \delta \quad \alpha \in (-1, 1].$$

Theorem 3 defines our sup-bound test, which maintains size over the nuisance parameter space, with $\bar{q}_{\hat{\rho},\delta}$ as the required sup-bound critical value. This sup-bound critical value can be easily found by simulation and, as a matter of fact, we found in our extensive simulation study that for a given δ and ρ , $q_{\rho,\delta}(c)$ is monotone decreasing in c , that is,

$$\bar{q}_{\rho,\delta} = \sup_{c \geq 0} \{q_{\rho,\delta}(c) : P(\Lambda_{c,\rho} > q_{\rho,\delta}(c)) = \delta\} = q_{\rho,\delta}(0).$$

This phenomenon of the sup-bound occurring at the unit root was also noticed in the sup-bound tests considered in Cavanagh, Elliot, and Stock (1995).

In general, the drawback of a sup-bound test is that the test can be too conservative if the null distribution changes a great deal over the nuisance parameter space (see Cavanagh, Elliot, and Stock 1995), resulting in significant power losses. However, Remark 3 above shows that the distributions of the QRLRT for different values of α look quite similar in the right tail. As a consequence, the sup-bound test for the QRLRT should not be expected to be very conservative. Table 1 below demonstrates this by computing the tail probability of $\Lambda_{c,\rho}$ beyond sup-bound critical values $\bar{q}_{\rho,0.05}$ for different values of c and ρ at $\delta = 5\%$. It is seen that though the sup-bound test can become conservative, it is not very undersized, suggesting that any power losses will not be significant. This is indeed found to be the case in our simulation study presented in Section 4.

In the next theorem, we establish the limiting distributions of the QRLRT statistic under the local alternatives $H_a : \beta = b/s_n$, where $s_n = \sqrt{n}$, $\sqrt{nk_n}$ and n for α that is fixed, deviates moderately from unity and is local to unity, respectively.

Theorem 4. For the QRLRT Λ_n , we have

(i) If α is fixed and $\beta = b/\sqrt{n}$ then

$$\Lambda_n \xrightarrow{D} \chi_1^2\{b^2/(1 - \alpha^2)\},$$

a noncentral χ_1^2 with noncentrality parameter $b^2/(1 - \alpha^2)$.

(ii) If $\alpha = 1 - c/k_n$, where $k_n = n^\lambda$, $\lambda \in (0, 1)$, and $\beta = b/\sqrt{nk_n}$, then

$$\Lambda_n \xrightarrow{D} \chi_1^2\{b^2/(2c)\}$$

a noncentral χ_1^2 with noncentrality parameter $b^2/(2c)$.

(iii) If $\alpha = 1 - c/n$ and $\beta = b/n$, then $\Lambda_n \xrightarrow{D} \Lambda_{c,\rho}(b)$, where

$$\Lambda_{c,\rho}(b) = \left\{ \rho g_{c,\rho}^{1/2} \tau_c + \sqrt{1 - \rho^2 g_{c,\rho}} \mathcal{Z} + b g_{c,\rho}^{1/2} \sqrt{\int J_c^2(r) dr} \right\}^2$$

where $\mathcal{Z}$, τ_c , and $g_{c,\rho}$ are defined as in Theorem 2.

3.1 The AR(ρ) Regressor Case

The above results in Theorem 2 for the QRLRT also extend easily to the more general case where the regressor series follows an AR(p) process. Most specifically, let

$$Y_t = \eta + \beta X_{t-1} + u_t \quad (3.3)$$

$$X_t = \mu + \alpha_1 X_{t-1} + \cdots + \alpha_p X_{t-p} + v_t.$$

Letting $\alpha = \sum_{i=1}^p \alpha_i$ and $\gamma = (\gamma_1, \dots, \gamma_{p-1})$, where $\gamma_s = -\sum_{j=s}^p \alpha_j$, we can rewrite X_t in the Dickey-Fuller form,

$$X_t = \mu + \alpha X_{t-1} + \gamma_1 \nabla X_{t-1} + \cdots + \gamma_{p-1} \nabla X_{t-p+1} + v_t.$$

The expressions of the WLSRL $Q_n(\beta, \alpha, \gamma)$ along with the WLSRL estimator $(\hat{\beta}, \hat{\alpha}_{\text{WLS}}, \hat{\gamma}'_{\text{WLS}})'$ and the constrained estimator under $H_0 : \beta = 0$, $(\hat{\alpha}_{\text{WLS}}^0, \hat{\gamma}'_{\text{WLS}}^0)'$ are given by Equations (A.1), (A.2), and (A.3) in Appendix A. The proof of the following theorem is similar to that of Theorem 2, except it is much more lengthier, we thus omit it.

Theorem 5. Let $Q_n(\beta, \alpha, \gamma)$ be the WLSRL objected function under model 3.3, $(\hat{\beta}, \hat{\alpha}_{\text{WLS}}, \hat{\gamma}'_{\text{WLS}})'$ be the WLSRL estimator and $(\hat{\alpha}_{\text{WLS}}^0, \hat{\gamma}'_{\text{WLS}}^0)'$ be the WLSRL estimator under constraint $\beta = 0$. Then the QRLRT

$$\Lambda_n = Q_n(0, \hat{\alpha}_{\text{WLS}}^0, \hat{\gamma}_{\text{WLS}}^0) - Q_n(\hat{\beta}_{\text{WLS}}, \hat{\alpha}_{\text{WLS}}, \hat{\gamma}_{\text{WLS}})$$

has the same limiting distributions as stated in Theorem 2.

We now turn toward studying the finite sample size and power of our procedure through simulations.

4. SIMULATION STUDY

In this section, we carry out Monte Carlo studies to compare the size and power of our sup-bound-based QRLRT test given

Table 2. Rejection rates for testing $H_0 : \beta = 0$. True $\beta = \frac{b\sqrt{1-\rho^2}}{n}$, $\rho = -0.98$, $\alpha = 1 - \frac{c}{n}$, nominal size = 5%, 5000 replications

c	n	$b = 0$		$b = 25$		$b = 50$	
		QRLRT	JM	QRLRT	JM	QRLRT	JM
0	100	0.0526	0.0462	0.9514	0.7718	1	0.9852
	200	0.0534	0.0544	0.9536	0.7846	1	0.9940
	400	0.0522	0.0448	0.9552	0.7906	1	0.9992
1	100	0.0472	0.0506	0.8832	0.4796	0.9998	0.9990
	200	0.0460	0.0522	0.8910	0.4688	1	0.9990
	400	0.0440	0.0500	0.8908	0.4680	1	0.9996
5	100	0.0400	0.0492	0.2556	0.0782	0.9872	0.5454
	200	0.0400	0.0500	0.2658	0.0828	0.9922	0.5352
	400	0.0394	0.0568	0.2620	0.0886	0.9908	0.5124
10	100	0.0416	0.0560	0.1598	0.0724	0.6682	0.1110
	200	0.0378	0.0500	0.1574	0.0650	0.6752	0.1016
	400	0.0396	0.0570	0.1512	0.0728	0.6684	0.1044
20	100	0.0460	0.0610	0.1182	0.0646	0.3428	0.0784
	200	0.0368	0.0570	0.1052	0.0634	0.3198	0.0770
	400	0.0380	0.0622	0.0994	0.0684	0.3138	0.0804

in Theorem 3 with the test proposed in Jansson and Moreira (2006) (henceforth, JM). The data were generated from

$$\begin{aligned} Y_t &= \beta X_{t-1} + u_t \\ X_t &= \alpha X_{t-1} + v_t \end{aligned} \quad \alpha = 1 - \frac{c}{n} \quad \begin{pmatrix} u_t \\ v_t \end{pmatrix} \stackrel{\text{iid}}{\sim} N\left(0, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}\right),$$

where the intercept in each equation was set to zero without loss of generality, since our test is location invariant. As mentioned earlier in Section 3, initial simulation results showed that for a given pair (ρ, δ) , the sup-bound critical value was $q_{\rho, \delta}(0)$, that is, the supremum, over c , of the critical value was attained at $c = 0$. Hence, for each replication, we first estimated ρ and then used $q_{\hat{\rho}, \delta}(0)$ as the critical value for our test. Furthermore, since Chen and Deo (2010) reported that the iterated WLSRL estimates, obtained by using the first step WLSRL estimates in the weight function W_n , have better finite sample behavior, we computed the QRLRT using these iterated WLSRL estimates.

In Table 2, we compare our test procedure with JM's conditional test for $\rho = -0.98$. This large negative value of ρ was chosen to reflect the kind of values observed empirically (see Chen and Deo 2006). Simulation studies not reported here for

other values of ρ yielded results similar in nature to those reported here for $\rho = -0.98$. The sample size n was set to be 100, 200, and 400 with $c = 0, 1, 5, 10$, and 20. Following JM's set-up, the parameter β was set to be $\frac{b\sqrt{1-\rho^2}}{n}$, where $b = 0, 25$, and 50. The rejection rates when $b = 0$ correspond to the size of the tests and when $b = 25$ and 50, to the power against local alternatives. Not surprisingly, it is seen from the table that the size of our test is slightly below the nominal level when c takes large value, since QRLRT is asymptotically conservative. However, despite the conservative size, the power of our test is uniformly higher than that of the JM test, with the power gains being very substantial at times.

In Table 3, we provide, for comparison, simulation results for the configuration presented in Table II on page 702 of JM. Following the settings in Table II of JM, the correlation ρ is set to -0.5 and 0.5 , while the sample size is set at $n = 1000$ with 500 replications. Once again, it is seen that the QRLRT maintains size while delivering power that is either identical to or higher than that of the JM test, with several power gains being very substantial.

Table 3. Rejection rates for testing $H_0 : \beta = 0$. True $\beta = \frac{b\sqrt{1-\rho^2}}{n}$, $\alpha = 1 - \frac{c}{n}$, nominal size = 5%, $n = 1000$, 500 replications

ρ	c	$c = 0$		$c = 5$		$c = 10$		$c = 15$	
		QRLRT	JM	QRLRT	JM	QRLRT	JM	QRLRT	JM
-0.5	0	0.040	0.054	0.042	0.058	0.056	0.042	0.046	0.066
	5	0.488	0.424	0.186	0.102	0.138	0.076	0.112	0.080
	10	0.892	0.826	0.620	0.338	0.438	0.180	0.316	0.154
	15	0.990	0.946	0.904	0.702	0.772	0.392	0.646	0.230
0.5	0	0.062	0.046	0.052	0.036	0.064	0.018	0.058	0.022
	5	0.398	0.412	0.196	0.132	0.150	0.054	0.114	0.040
	10	0.728	0.602	0.504	0.220	0.372	0.122	0.300	0.088
	15	0.908	0.720	0.746	0.274	0.622	0.196	0.508	0.082

APPENDIX A: MODEL WITH AN AR (p) REGRESSOR

We start by defining some notation. For $t = p + 1, \dots, n$ and any time series ζ_t , we will denote the corresponding sample mean corrected series and its lag series as,

$$\begin{aligned}\zeta_{t,\hat{\mu}} &= \zeta_t - \frac{1}{n-p} \sum_{u=p+1}^n \zeta_u, \\ \zeta_{t-1,\hat{\mu}} &= \zeta_{t-1} - \frac{1}{n-p} \sum_{u=p+1}^n \zeta_{u-1}, \\ \nabla \zeta_{t-k,\hat{\mu}} &= \nabla \zeta_{t-k} - \frac{1}{n-p} \sum_{u=p+1}^n \nabla \zeta_{u-k}.\end{aligned}$$

For $t = p + 1, \dots, n$, let

$$\begin{aligned}\epsilon_{t,\hat{\mu}}(\beta, \alpha, b) &= \begin{pmatrix} u_{t,\hat{\mu}} \\ v_{t,\hat{\mu}} \end{pmatrix} \\ &= \begin{pmatrix} Y_{t,\hat{\mu}} - \beta X_{t-1,\hat{\mu}} \\ X_{t,\hat{\mu}} - \alpha X_{t-1,\hat{\mu}} - \sum_{k=1}^{p-1} \gamma_k \nabla X_{t-k,\hat{\mu}} \end{pmatrix}.\end{aligned}$$

Furthermore, denote the series with the first observation subtracted as, $\zeta_{s,d} = \zeta_s - \zeta_1$, $s = 1, \dots, n$, as before. Define $R(\beta, \alpha, \gamma) = (R_1, \dots, R_p)$, where

$$\begin{aligned}R_1 &= \begin{pmatrix} Y_{2,d} - \beta X_{1,d} \\ X_{2,d} - \alpha X_{1,d} \end{pmatrix} \\ R_s &= \begin{pmatrix} Y_{s+1,d} - \beta X_{s,d} \\ X_{s+1,d} - \alpha X_{s,d} - \sum_{k=1}^{s-1} \gamma_k \nabla X_{s-k} \end{pmatrix}, \quad s = 2, \dots, p-1,\end{aligned}$$

and

$$\begin{aligned}R_p &= \frac{1}{\sqrt{n-p}} \sum_{t=p+1}^n r_t \quad \text{where} \\ r_t &= \begin{pmatrix} Y_{t,d} - \beta X_{t-1,d} \\ X_{t,d} - \alpha X_{t-1,d} - \sum_{k=1}^{p-1} \gamma_k \nabla X_{t-k} \end{pmatrix}.\end{aligned}$$

Let $(\hat{\Sigma}, \hat{\beta}, \hat{\alpha}, \hat{\gamma})$ be any consistent estimators of $(\Sigma, \beta, \alpha, \gamma)$. From Lemma 1 of Chen and Deo (2010), the WLSRL of model (3.3) based on the observations $(Y_t, X_t)'$, $t = 1, \dots, n$ is given by

$$Q_n(\beta, \alpha, \gamma) = Q_{OLS}(\beta, \alpha, \gamma) + Q_W(\beta, \alpha, \gamma), \quad (\text{A.1})$$

where

$$\begin{aligned}Q_{OLS}(\beta, \alpha, \gamma) &= \sum_{t=2}^n \epsilon'_{t,\hat{\mu}}(\beta, \alpha, \gamma) \hat{\Sigma}^{-1} \epsilon_{t,\hat{\mu}}(\beta, \alpha, \gamma), \\ Q_W(\beta, \alpha, \gamma) &= \text{vec}\{R(\beta, \alpha, \gamma)\}' \hat{W}_n \text{vec}\{R(\beta, \alpha, \gamma)\},\end{aligned}$$

and $\hat{W}_n = W_n(\hat{\Sigma}, \hat{\beta}, \hat{\alpha}, \hat{\gamma})$ with $W_n(\Sigma, \beta, \alpha, \gamma) = \{I_p \otimes \Sigma + D' \Sigma D\}^{-1}$, where

$$\begin{aligned}D &= ((I - H_1 - H_2)' \quad \dots \quad (I - H_1 - H_p)' \\ &\quad \times \sqrt{n-p}(I - H_1)')\end{aligned}$$

and

$$H_1 = \begin{pmatrix} 0 & \beta \\ 0 & \alpha \end{pmatrix}, \quad H_s = \begin{pmatrix} 0 & 0 \\ 0 & \gamma_{s-1} \end{pmatrix}, \quad s = 2, \dots, p.$$

For the WLSRL estimator of (β, α, γ) , define

$$\begin{aligned}V_Z &= \left(Z_{2,d}, \dots, Z_{p,d}, \frac{1}{\sqrt{n-p}} \sum_{t=p+1}^n Z_{t,d} \right), \\ U &= \left(X_{1,d}, \dots, X_{p-1,d}, \frac{1}{\sqrt{n-p}} \sum_{t=p+1}^n X_{t-1,d} \right),\end{aligned}$$

and the $(p-1) \times p$ matrix $S = (S_1, \dots, S_p)$, where

$$S'_k = (\nabla X_s, \dots, \nabla X_1, 0, \dots, 0), \quad k = 1, \dots, p-1$$

$$S'_p = \frac{1}{\sqrt{n-p}} \sum_{t=p+1}^n (\nabla X_{t-1}, \dots, \nabla X_{t-p+1}).$$

Further define the $p \times 1$ vector $L_t = (\nabla X_{t-1,\hat{\mu}}, \dots, \nabla X_{t-p+1,\hat{\mu}})'$. Let A_{r2} , A_{c2} , A_{22} denote the second row, the second column, and the $(2, 2)$ entry of a 2×2 matrix A , respectively, the WLSRL estimator

$$\begin{aligned}(\hat{\beta}_{WLS}, \hat{\beta}'_{WLS}, \hat{\gamma}'_{WLS})' &= \left\{ 2 \sum_{t=p+1}^n \begin{pmatrix} X_{t-1}^2 \hat{\Sigma}^{-1} & X_{t-1} L'_t \otimes \hat{\Sigma}_{c2}^{-1} \\ X_{t-1} L_t \otimes \hat{\Sigma}_{r2}^{-1} & L_t L'_t \hat{\Sigma}_{22}^{-1} \end{pmatrix} \right. \\ &\quad \left. + \sum_{i,j=1}^p \begin{pmatrix} 2U_i U_j \hat{W}^{ij} & (U_i S'_j + U_j S'_i) \otimes \hat{W}_{c2}^{ij} \\ (U_i S'_j + U_j S'_i) \otimes \hat{W}_{r2}^{ij} & (S_i S'_j + S_j S'_i) \hat{W}_{22}^{ij} \end{pmatrix} \right\}^{-1} \\ &\quad \times \left\{ 2 \sum_{t=p+1}^n \begin{pmatrix} \hat{\Sigma}^{-1} Z_{t,\hat{\mu}} X_{t-1,\hat{\mu}} \\ \hat{\Sigma}_{r2}^{-1} Z_{t,\hat{\mu}} L_t \end{pmatrix} \right. \\ &\quad \left. + \sum_{i,j=1}^p \begin{pmatrix} \hat{W}^{ij} (V_{Z,i} U_j + V_{Z,j} U_i) \\ [\hat{W}_{r2}^{ij} (V_{Z,i} S'_j + V_{Z,j} S'_i)]' \end{pmatrix} \right\}, \quad (\text{A.2})\end{aligned}$$

where $\hat{W}^{ij}$ is the (i, j) block, a 2×2 matrix, of $\hat{W}_n$ and $V_{Z,i}$ is the i th column of V_Z . Let

$$U_X = \begin{pmatrix} U \\ S \end{pmatrix} = (U_{X,1}, \dots, U_{X,p}) \quad \text{and} \quad L_{X,t} = \begin{pmatrix} X_{t-1,\hat{\mu}} \\ L_t \end{pmatrix},$$

the WLSRL estimator under $H_0 : \beta_0$ is given by

$$\begin{pmatrix} \hat{\alpha}_{\text{WLS}}^0 \\ \hat{\gamma}_{\text{WLS}}^0 \end{pmatrix} = \left\{ 2 \sum_{t=p+1}^n L_{X,t} L'_{X,t} \hat{\Sigma}_{22} + \sum_{i,j=1}^p (U_{X,i} U'_{X,j} + U_{X,j} U'_{X,i}) \hat{W}_{22}^{ij} \right\}^{-1} \times \left\{ 2 \sum_{t=p+1}^n \hat{\Sigma}_{r2}^{-1} Z_{t,\hat{\mu}} L_{X,t} + \sum_{i,j=1}^p [\hat{W}_{r2}^{ij} (V_{Z,i} U'_{X,j} + V_{Z,j} U'_{X,i})]' \right\}. \quad (\text{A.3})$$

APPENDIX B: PROOF

Throughout this appendix, we will use the summation notation $\sum_t$ for $\sum_{t=2}^n$ unless stated otherwise.

Proof of Theorem 1. Without loss of generality, we will prove for $\beta = 0$. Following from Phillips (1987), Phillips and Magdalinos (2007), and Giraitis and Phillips (2012),

$$\sum_t X_{t-1,\hat{\mu}}^2 = O_p(nk_n), \quad \sum_t X_{t-1,d} = O_p(k_n \sqrt{n} + n). \quad (\text{B.1})$$

Using the matrix inversion formulas and letting

$$\eta_n = \frac{\det(\hat{\Sigma} \hat{W}_n) (\sum_s X_{s-1,d})^2}{n-1 \sum_t X_{t-1,\hat{\mu}}^2} = O_p\left(\frac{k_n}{n^2}\right),$$

by (B.1), Lemma 2, and Corollary 2, the WLS estimate in (2.3) becomes

$$\begin{pmatrix} \hat{\beta}_{\text{WLS}} \\ \hat{\alpha}_{\text{WLS}} \end{pmatrix} = \frac{1}{\Delta_n} \{I + \eta_n \hat{W}_n^{-1} \hat{\Sigma}^{-1}\} \times \left(\sum_t X_{t-1,\hat{\mu}} Z_{t,\hat{\mu}} + \frac{\hat{\Sigma} \hat{W}_n}{n-1} \sum_s \sum_t X_{s-1,d} Z_{t,d} \right),$$

where

$$\Delta_n = \sum_t X_{t-1,\hat{\mu}}^2 \left\{ 1 + \frac{1}{n-1} (\sum_s X_{s-1,d})^2 \sum_t X_{t-1,\hat{\mu}}^2 \times (\hat{w}_{11} + \hat{w}_{22} + 2\hat{\rho}\hat{w}_{12} + \eta_n) \right\}.$$

With detailed algebraic calculation and writing $u_t = \rho v_t + e_t$,

$$(\hat{\beta}_{\text{WLS}}, \hat{\alpha}_{\text{WLS}} - \alpha_0)' = (\mathcal{S}_n + \hat{\rho} \mathcal{T}_n, \mathcal{T}_n)' + R'_n, \quad (\text{B.2})$$

where

$$\mathcal{S}_n = \frac{\sum_t X_{t-1,\hat{\mu}} e_{t,\hat{\mu}}}{\sum_t X_{t-1,\hat{\mu}}^2},$$

$$\mathcal{T}_n = \frac{\sum_t X_{t-1,\hat{\mu}} v_{t,v} + \frac{\hat{w}_{22}}{n-1} \sum_{s,t} X_{s-1,d} (X_{t,d} - \alpha_0 X_{t-1,d})}{\sum_t X_{t-1,\hat{\mu}}^2 + \frac{\hat{w}_{22}}{n-1} (\sum_s X_{s-1,d})^2},$$

and the remainder term $R_n = O_p(1/n)$ if $\lambda \in [0, 1/2]$, and $R_n = O_p(1/k_n \sqrt{n})$ if $\lambda \in [1/2, 1]$ by (B.1), Lemmas 2 and Corollary 2. The theorem follows from Giraitis and Phillips (2012), the theorem follows from Phillips (1987) and Phillips and Magdalinos (2007). $\square$

Proof of Lemma 1. The restricted estimate of α in (3.1) is

$$\hat{\alpha}_{\text{WLS}}^0 = \alpha_0 + (1 - \rho^2) \mathcal{T}_n^0 + \hat{\rho} \mathcal{S}_n^0 + R_n^0,$$

where

$$\mathcal{T}_n^0 = \frac{\sum_t X_{t-1,\hat{\mu}} v_{t,v} + \frac{\hat{w}_{22}}{n-1} \sum_{s,t} X_{s-1,d} (X_{t,d} - \alpha_0 X_{t-1,d})}{\sum_t X_{t-1,\hat{\mu}}^2 + \frac{(1-\hat{\rho}^2)\hat{w}_{22}}{n-1} (\sum_s X_{s-1,d})^2},$$

$$\mathcal{S}_n^0 = \frac{\sum_t X_{t-1,\hat{\mu}} e_{t,\hat{\mu}}}{\sum_t X_{t-1,\hat{\mu}}^2 + \frac{(1-\hat{\rho}^2)\hat{w}_{22}}{n-1} (\sum_s X_{s-1,d})^2}$$

and

$$R_n^0 = \frac{\frac{\hat{w}_{12}}{n-1} \sum_s X_{s-1,d} \sum_t Y_{t,d}}{\sum_t X_{t-1,\hat{\mu}}^2 + \frac{(1-\hat{\rho}^2)\hat{w}_{22}}{n-1} (\sum_s X_{s-1,d})^2}.$$

Let $\mathcal{S}_n$ and $\mathcal{T}_n$ be the same as in (B.2), and define

$$\begin{aligned} g_n &:= \frac{\sum_t X_{t-1,\hat{\mu}}^2}{\sum_t X_{t-1,\hat{\mu}}^2 + \frac{(1-\hat{\rho}^2)\hat{w}_{22}}{n-1} (\sum_s X_{s-1,d})^2} \\ &= \begin{cases} 1 + O_p(k_n/n), & \lambda \in [0, 1) \\ g_{\rho,c} + O_p(n^{-3/2}), & \lambda = 1 \end{cases}, \end{aligned} \quad (\text{B.3})$$

by (B.1), Lemma 2 and Corollary 2. We obtain

$$\hat{\alpha}_{\text{WLS}}^0 - \alpha_0 = (1 - \hat{\rho}^2 g_n) \mathcal{T}_n + \hat{\rho} g_n \mathcal{S}_n + R_n^0. \quad (\text{B.4})$$

The remainder term $R_n^0 = O_p(1/n)$ if $\lambda \in [0, 1/2]$ and $R_n^0 = O_p(1/k_n \sqrt{n})$ if $\lambda \in [1/2, 1]$. The limiting distribution of $(\hat{\alpha}_{\text{WLS}}^0 - \alpha_0)$ follows from (B.4) and Lemma 3. $\square$

Proof of Theorem 2. We will prove only for Λ_Q , the proof of $\Lambda_{\mathcal{L}}$ will follow by the same proof in combination with Lemma 2 and Corollary 2. For algebraic convenience, we use the following notations:

$$\mathcal{Q}_n = \sum_t X_{t-1,\hat{\mu}}^2, \quad \text{and} \quad \mathcal{S}_n = \frac{1}{\sqrt{n-1}} \sum_s X_{s-1,d}.$$

The second partial derivatives of $Q_n(\beta, \alpha)$ are

$$\begin{aligned} \frac{\partial^2 Q_n(\beta, \alpha)}{\partial \beta^2} &= \frac{2\mathcal{Q}_n}{1 - \hat{\rho}^2} + 2\hat{w}_{11} \mathcal{S}_n^2 = \frac{2\mathcal{Q}_n}{1 - \hat{\rho}^2} + O_p(1), \\ \frac{\partial^2 Q_n(\beta, \alpha)}{\partial \alpha \partial \beta} &= -\frac{2\hat{\rho} \mathcal{Q}_n}{1 - \hat{\rho}^2} + 2\hat{w}_{12} \mathcal{S}_n^2 = -\frac{2\hat{\rho} \mathcal{Q}_n}{1 - \hat{\rho}^2} + O_p(k_n), \\ \frac{\partial^2 Q_n(\beta, \alpha)}{\partial \alpha^2} &= \frac{2\mathcal{Q}_n}{1 - \hat{\rho}^2} + 2\hat{w}_{22} \mathcal{S}_n^2, \end{aligned}$$

by (B.1), Lemma 2 and Corollary 2. With a Taylor expansion of Q_n ,

$$\begin{aligned} \Lambda_n &= Q_n(0, \hat{\alpha}_{\text{WLS}}^0) - Q_n(\hat{\beta}_{\text{WLS}}, \hat{\alpha}_{\text{WLS}}) \\ &= \frac{1}{2} \left\{ \frac{\partial^2 Q_n(\hat{\beta}_{\text{WLS}}, \hat{\alpha}_{\text{WLS}})}{\partial \beta^2} (\beta_0 - \hat{\beta}_{\text{WLS}})^2 \right. \\ &\quad + \frac{2\partial^2 Q_n(\hat{\beta}_{\text{WLS}}, \hat{\alpha}_{\text{WLS}})}{\partial \alpha \partial \beta} (\beta_0 - \hat{\beta}_{\text{WLS}})(\hat{\alpha}_{\text{WLS}}^0 - \hat{\alpha}_{\text{WLS}}) \\ &\quad \left. + \frac{\partial^2 Q_n(\hat{\beta}_{\text{WLS}}, \hat{\alpha}_{\text{WLS}})}{\partial \alpha^2} (\hat{\alpha}_{\text{WLS}}^0 - \hat{\alpha}_{\text{WLS}})^2 \right\}. \end{aligned} \quad (\text{B.5})$$

Writing $\hat{\alpha}_{\text{WLS}}^0 - \hat{\alpha}_{\text{WLS}} = (\hat{\alpha}_{\text{WLS}}^0 - \alpha_0) - (\hat{\alpha}_{\text{WLS}} - \alpha_0)$ and using (B.2) and (B.4), Equation (B.5) becomes

$$\begin{aligned} & \frac{\varrho_n}{1 - \hat{\rho}^2} \{S_n + \hat{\rho}T_n\}^2 - \frac{2\hat{\rho}\varrho_n}{1 - \hat{\rho}^2} \{S_n + \hat{\rho}T_n\}(-\hat{\rho}^2 g_n T_n + \hat{\rho} g_n S_n) \\ & + \frac{\varrho_n + (1 - \hat{\rho}^2)\hat{w}_{22}S_n^2}{1 - \hat{\rho}^2} (-\hat{\rho}^2 g_n T_n + \hat{\rho} g_n S_n)^2 \\ & = \left\{ \varrho_n + \hat{w}_{22}S_n^2 \right\} \hat{\rho}^2 g_n T_n^2 + \frac{(1 - \hat{\rho}^2 g_n)\varrho_n}{1 - \hat{\rho}^2} S_n^2 \\ & + \frac{2(1 - \hat{\rho}^2 g_n)\hat{\rho}\varrho_n}{1 - \hat{\rho}^2} S_n T_n \\ & = \left\{ \sqrt{\varrho_n + \hat{w}_{22}S_n^2} \hat{\rho} g_n^{1/2} T_n + \sqrt{\frac{(1 - \hat{\rho}^2 g_n)\varrho_n}{1 - \hat{\rho}^2}} S_n \right\}^2. \end{aligned}$$

The limiting distribution of Λ_n follows from Lemmas 4 and 5. $\square$

Proof of Theorem 4. Let $\beta_n = b/\sqrt{nk_n}$ and $\hat{\alpha}_\beta$ be the WLSRL estimator of α for fixed β . Now,

$$\begin{aligned} 0 &= \frac{\partial Q_n(\beta_0, \hat{\alpha}, \beta_0)}{\partial \alpha} - \frac{\partial Q_n(\beta_n, \hat{\alpha}, \beta_n)}{\partial \alpha} \\ &= \frac{\partial^2 Q_n}{\partial \alpha \partial \beta}(\beta_0 - \beta_n) + \frac{\partial^2 Q_n}{\partial \alpha^2}(\hat{\alpha}_{\beta_0} - \hat{\alpha}_{\beta_n}) + o_p(\sqrt{nk_n}). \end{aligned}$$

We get, from the above equation,

$$\begin{aligned} \hat{\alpha}_{\beta_0} - \hat{\alpha}_{\beta_n} &= -\frac{b}{\sqrt{nk_n}} \frac{\partial^2 Q_n}{\partial \alpha \partial \beta} \left\{ \frac{\partial^2 Q_n}{\partial \alpha^2} \right\}^{-1} \\ &= -\frac{b}{\sqrt{nk_n}} \hat{\rho} g_n + o_p(1/\sqrt{nk_n}). \end{aligned}$$

Under $H_A : \beta = \beta_n$, the expressions of (B.2) and (B.4) are valid with β_0 being replaced by β_n , thus, together with the above equation,

$$\begin{aligned} \hat{\alpha}_{\text{WLS}}^0 - \hat{\alpha}_{\text{WLS}} &= \hat{\alpha}_{\beta_0} - \hat{\alpha}_{\beta_n} + \hat{\alpha}_0 - \alpha_0 - (\hat{\alpha}_{\text{WLS}} - \alpha_0) \\ &= -\hat{\rho}^2 g_n T_n + \hat{\rho} g_n S_n - \frac{\hat{\rho} g_n b}{\sqrt{nk_n}} + o_p(1/\sqrt{nk_n}) \\ \beta_0 - \hat{\beta}_{\text{WLS}} &= \beta_0 - \beta_n - (\hat{\beta}_{\text{WLS}} - \beta_n) = -\frac{b}{\sqrt{nk_n}} - (S_n + \hat{\rho}T_n). \end{aligned}$$

Plugging the above equations into (B.5) and ignoring the smaller order terms,

$$\begin{aligned} \Lambda_n &= \left\{ \frac{\rho \varrho_n^{1/2} \sqrt{1 - \hat{\rho}^2 g_n}}{\sqrt{1 - \hat{\rho}^2}} T_n + \sqrt{\frac{(1 - \hat{\rho}^2 g_n)\varrho_n}{1 - \hat{\rho}^2}} \left(S_n + \frac{b}{\sqrt{nk_n}} \right) \right\}^2 \\ &= \left\{ \hat{\rho} g_n^{1/2} (\varrho_n + w_{22}S_n^2)^{1/2} T_n + \sqrt{\frac{(1 - \hat{\rho}^2 g_n)\varrho_n}{1 - \hat{\rho}^2}} S_n \right. \\ & \quad \left. + \frac{g_n^{1/2} (\varrho_n + w_{22}S_n^2)^{1/2} b}{\sqrt{nk_n}} \right\}^2, \end{aligned}$$

because

$$\sqrt{\frac{(1 - \hat{\rho}^2 g_n)\varrho_n}{1 - \hat{\rho}^2}} = g_n^{1/2} (\varrho_n + w_{22}S_n^2)^{1/2}.$$

The theorem follows from Lemmas 4 and 5. $\square$

Lemma 2. Let $W_n(\rho, \beta, \alpha)$ be defined as (Corollary 2.2) and $\alpha = 1 - c/n^\lambda$, $\lambda \in [0, 1]$. If $\beta = 0$, then

$$\det(W_n) = \begin{cases} O(n^{-2+2\lambda}), & \lambda \in [0, 1/2] \\ (n-1)^{-1} + O(n^{-2\lambda}), & \lambda \in (1/2, 1] \end{cases}$$

and furthermore,

$$\begin{aligned} W_n(\rho, 0, \alpha) &= \begin{cases} n^{-1+2\lambda} \{c^2(1 - \rho^2)\}^{-1} \Xi + O(n^{-1+\lambda}), & \lambda \in [0, 1/2] \\ \{1 + c^2(1 - \rho^2)\}^{-1} \Xi + O(n^{-1/2}), & \lambda = 1/2 \\ \Xi + O(n^{-\lambda}), & \lambda \in (1/2, 1] \end{cases}, \\ \text{where } \Xi &= \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

Proof. With a few algebraic steps,

$$\begin{aligned} W_n(\rho, \beta, \alpha) &= \frac{1}{\det \{W_n^{-1}(\rho, \beta, \alpha)\}} \\ &\times \begin{bmatrix} 1 + (n-1)(1 - \alpha)^2 & -\rho - (n-1)\{(1 - \alpha)(\rho - \beta)\} \\ -\rho - (n-1) & 1 + (n-1)(1 - 2\beta\rho + \beta^2) \\ \{(1 - \alpha)(\rho - \beta)\} & \end{bmatrix} \end{aligned}$$

and the determinant of $W_n^{-1}(\rho, \beta, \alpha)$,

$$\begin{aligned} \det \{W_n^{-1}(\rho, \beta, \alpha)\} &= (n-1)\{(1 + 2\rho\beta + \beta^2) - 2\rho(\beta + \rho)(1 - \alpha) \\ & \quad + (1 - \alpha)^2\} + (1 - \rho^2)\{1 + (n-1)^2(1 - \alpha)^2\} \end{aligned}$$

We have, when $\beta = 0$,

$$\begin{aligned} \det \{W_n^{-1}(\rho, 0, \alpha)\} &= (n-1)\{(1 - 2\rho^2(1 - \alpha) + (1 - \alpha)^2) \\ & \quad + (1 - \rho^2)\{1 + (n-1)^2(1 - \alpha)^2\} \\ &= \begin{cases} (n-1)^{2-2\lambda} c^2(1 - \rho^2) + O(n), & \alpha = 1 - c/n^\lambda, \\ & \lambda \in [0, 1/2] \\ (n-1)\{1 + c^2(1 - \rho^2)\}, & \alpha = 1 - c/\sqrt{n}, \\ (n-1)(1 + O(n^{1-2\lambda})), & \alpha = 1 - c/n^\lambda, \\ & \lambda \in (1/2, 1] \end{cases} \end{aligned}$$

and,

$$W_n(\rho, 0, \alpha) = \frac{1}{\det \{W_n^{-1}(\rho, 0, \alpha)\}} \begin{pmatrix} 1 + c^2/n & -\rho(1 + c) \\ -\rho(1 + c) & n \end{pmatrix}.$$

The lemma follows immediately. $\square$

Corollary 1. Let $W_n(\rho, \beta, \alpha)$ be defined as 2.2, then the expressions of W_n in Lemma 2 hold for any value of β .

Corollary 2. Let $\hat{\rho}$ be a consistent estimate of ρ and $\hat{\beta}$ be a consistent estimate of β . Furthermore, let $\hat{\alpha}$ be a consistent estimate of α such that $n^{(1+\lambda)/2}|\hat{\alpha} - \alpha| = O_p(1)$ where $\alpha = 1 - c/n^\lambda$, $\lambda \in [0, 1]$. Then the expression of W_n in Lemma 2 holds for $W_n(\hat{\rho}, \hat{\beta}, \hat{\alpha})$ holds with the order of remainder terms $O(\cdot)$ being replaced by $O_p(\cdot)$. $\square$

Lemma 3. Let w_{22} be the (2, 2)th entry of $W_n(\alpha, \beta, \rho)$ in (2.2),

$$\begin{aligned} & \sum_t X_{t-1,\hat{\mu}} v_{t,\hat{\mu}} + \frac{w_{22} \left\{ \sum_{s,t} X_{s-1,d} X_{t,d} - \alpha \left(\sum_s X_{s-1,d} \right)^2 \right\}}{n-1} \\ &= \begin{cases} \sum_t X_{t-1,\hat{\mu}} v_{t,\hat{\mu}} + O_p(n^\lambda), & \lambda \in [0, 1/2] \\ \sum_t X_{t-1} v_t + O_p(\sqrt{n}), & \lambda \in [1/2, 1] \end{cases} \end{aligned}$$

and

$$\begin{aligned} & \sum_t X_{t-1,\hat{\mu}}^2 + \frac{w_{22} \left(\sum_s X_{s-1,d} \right)^2}{n-1} \\ &= \begin{cases} \sum_t X_{t-1,\hat{\mu}}^2 + O_p(n^{2\lambda}), & \lambda \in [0, 1/2] \\ \sum_t X_{t-1}^2 + O_p(n^{1/2+\lambda}), & \lambda \in [1/2, 1] \end{cases}. \end{aligned}$$

Proof. The lemma follows from Lemma 2 that $w_{22} = O(n^{-1+2\lambda})$, when $\lambda \in [0, 1/2]$ and $w_{22} = 1 + o(1)$, when $\lambda \in (1/2, 1]$, and the fact that

$$\begin{aligned} & \sum_{s,t} X_{s-1,d} X_{t,d} - \alpha \left(\sum_s X_{s-1,d} \right)^2 \\ &= \sum_s (X_s - X_1) \sum_t \{v_t - (1 - \alpha)X_1\} \end{aligned}$$

and

$$\begin{aligned} & \left(\sum_s X_{s-1,d} \right)^2 = \left(\sum_s X_{s-1} \right)^2 - (n-1)X_1 \\ & \times \sum_s X_{s-1} + (n-1)^2 X_1^2. \end{aligned}$$

Lemma 4. Let $e_t = u_t - \rho v_t$ and

$$S_n = \frac{\sum_t X_{t-1,\hat{\mu}} e_{t,\hat{\mu}}}{\sum_t X_{t-1,\hat{\mu}}^2},$$

then

$$\sqrt{nk_n} S_n \xrightarrow{D} \begin{cases} N(0, (1 - \rho^2)(1 - \alpha^2)), & \alpha \text{ is fixed, } k_n = 1 \\ N(0, 2c(1 - \rho^2)), & \alpha = 1 - c/k_n, k_n = n^\lambda, \lambda \in (0, 1) \\ \left(\int J_{c,\mu}^2(r) dr \right)^{-1/2} \sqrt{1 - \rho^2} \mathcal{Z}, & \alpha = 1 - c/n. \end{cases}$$

Proof. Since $e_{t,\hat{\mu}}$ is independent of $X_{t-1,\hat{\mu}}$,

$$\frac{\sum_t X_{t-1,\hat{\mu}} e_{t,\hat{\mu}}}{\sqrt{(1 - \rho^2) \sum_t X_{t-1,\hat{\mu}}^2}} \Big| X_{t-1,\hat{\mu}} \Rightarrow \mathcal{Z}.$$

When α is fixed, $n^{-1} \sum_t X_{t-1,\hat{\mu}}^2 \rightarrow_p (1 - \alpha^2)^{-1}$. When $\alpha = 1 - c/n^\lambda$ with $\lambda \in (0, 1)$, $(nk_n)^{-1} \sum_t X_{t-1,\hat{\mu}}^2 \rightarrow_p 1/2c$ from Phillips and Magdalinos (2007), and $(n-1)^{-1} \sum_t X_{t-1,\hat{\mu}} = O_p(k_n^2/n)$ from Giraitis and Phillips (2012), thus

$$\frac{1}{nk_n} \sum_t X_{t-1,\hat{\mu}}^2 = \frac{1}{nk_n} \sum_t X_{t-1}^2 (1 + o_p(1)) \rightarrow_p \frac{1}{2c}.$$

Furthermore, when $\alpha = 1 - c/n$, $n^{-2} \sum_t X_{t-1,\hat{\mu}}^2 \Rightarrow \int J_{c\mu}(r)^2 dr$. $\square$

Lemma 5. Let w_{22} be the (2, 2)th entry of W_n , and

$$\mathcal{T}_n = \frac{\sum_t X_{t-1,\hat{\mu}} v_{t,v} + \frac{w_{22}}{n-1} \sum_{s,t} X_{s-1,d} (X_{t,d} - \alpha_0 X_{t-1,d})}{\sum_t X_{t-1,\hat{\mu}}^2 + \frac{w_{22}}{n-1} \left(\sum_s X_{s-1,d} \right)^2},$$

then

$$\begin{aligned} & \sqrt{nk_n} \mathcal{T}_n \xrightarrow{D} \\ & \times \begin{cases} N(0, (1 - \alpha^2)), & \alpha \text{ is fixed, } k_n = 1 \\ N(0, 2c), & \alpha = 1 - c/k_n, k_n = n^\lambda, \lambda \in (0, 1) \\ \sqrt{\int J_c(r)^2 dr} \tau_c, & \alpha = 1 - c/n. \end{cases} \end{aligned}$$

Proof. The lemma follows from Lemma 3, Phillips (1987), and Phillips and Magdalinos (2007). $\square$

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